CE/CS/SE 3354 · Software Engineering · Fall 2026
Project Milestone P2

Analysis and High-Level Design

Whoosh Events

Team	Team 05, section .003
Document version	1.0 (baseline for P2; P1 was version 0.1)
Date	November 8, 2026
Repository URL	https://github.com/AO2-Git/CS3354-Team05-Project
Git tag for this milestone	p2
Submitted on Canvas by	Austin Ou Yang

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HOW TO USE THIS TEMPLATE
This is not a new document. P2 continues the document you baselined at P1. Copy your P1 sections 1 to 3 into this file (updated under change control), then fill in the new sections 4 to 9. Your P1 project plan, open questions and baseline move to the end and are renumbered:
P1 section 4 (project plan) becomes section 10. P1 section 5 (open questions) becomes 11. P1 section 6 (baseline) becomes 12.
Delete every guidance, example and warning box (blue, amber and red) and the two marking pages at the end before you submit. They are instructions, not content. Boxes or placeholders such as "[TBD]", "PR #[n]" or "[Member 3]" left in the submitted document lose document-quality points.
Length: sections 4 to 9 should add roughly 12 to 18 pages, diagrams included. Every section below carries a budget. A longer document is not a better one.
Diagrams: draw them in any tool (draw.io, Lucid, PlantUML, or a tidy hand drawing that is scanned). Export each as an image for this document and commit the source file to /docs/p2/diagrams. A diagram that cannot be read cannot be credited for its content.
Required structure: using this file is optional, but its section numbering is required, because the rubric refers to it.
Rubric: the marking pages at the end show where the 60 points sit. Read them before you divide the work.

WHAT P2 DOES NOT ASK FOR (DELETE THIS BOX BEFORE SUBMITTING)
P2 is analysis and high-level design. At this milestone we are NOT asking for:
design class diagrams with operations, parameter types or visibility, and design-level sequence diagrams with operation signatures (Chapter 9 detailed design; they enter the document during P3 work)
production code (the only code in P2 is the small architecture spike in section 8)
a test plan (P3, after Chapter 11)
screen mock-ups or a visual design (not graded at this milestone; add them only if they settle an open question)
design patterns from the Gang of Four catalog, ATAM or utility trees (not required; do not spend your time there)
Work placed in the wrong milestone is not credited. Spend the time on sections 4 to 9.

0 Revision History, Contribution Record and Changes Since P1

All three tables below are graded. The contribution record carries the individual 20% of this milestone and must agree with your repository history.

Version	Date	Author	Summary of change
0.1	10/18/2026	Team 05	P1 baseline: proposal and requirements
1.0	11/08/2026	Team 05	P2 baseline: analysis and high-level design added

0.1 Individual contribution record (P2)

► OWNER: everyone writes their own row with real commit/PR links · due Sat Nov 7, 6 PM (delete this line before submitting)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Austin Ou Yang	Section 6.3, ADR-03, 6.3 Box Diagram, GitHub docs folder, Section 6.4, 6.4 Diagram	PR#4, Commits 4a17667, e8ad637, 8cc58ab, e6e95ca	
Habib Mashhood			
Jaiom Rola			
Ahmed Fouad	0.2 changes CH-01 to CH-04; P1 fixes in 2.2 (student poll of 7), 3.1 to 3.5 (FR-02, FR-04, QR-02, use case diagram, actor-goal list); 4.1 to 4.3 (UC-01, UC-02, UC-03, self-review log); 5.1 domain model; 5.2.1 and 5.2.2 scenario tables and sequence diagrams	Issue #1; PR #2 (branch p2/ahmed-analysis-models); commits ee07f0d (diagram sources, docs/p2/diagrams), 27d6550 (section text, docs/p2/sections), 83927f7 (P1 fixes in 1.1 to 2.6, context diagram) and 31fa6c3 (sequence diagram notation)	
Atharva Thombare	Section 5.3: Cross checked analysis sequence diagrams with the UC tables in 4.2. Section 9.1, verified the requirements for	Commits: 91541ac, 95e1abe, 7c293ed, 9605a70 PR #9	PR #2

	traceability with their UC, components and QAS/ADR. 9.2, Gap analysis table. Reviewed and commented feedback on PR #2		
Sinan Can Demir	Sections 6.5 (analysis objects to components), 6.6 (design principles applied), 7.2 (trade-off summary), 10.2 (plan to P3) and 10.5 (AI-use declaration, collected from the team). Collected the signatures for section 12. Reviewed teammates' work: S-01 spike (PR #5) and the analysis models (PR #2).	PR #6, branch p2/sinan-summary-tables, commit d0935fd (draft of the sections above), 01ca14e and cd4ccbd. Review comments on PR #5 and a comment on PR #2.	PR #5 PR #2

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Evidence means something a reader can open and that existed before the deadline: a commit hash, a pull request, a file in the repository. Issues you own count as indirect evidence; link the commits that close them for full credit.
Commit your own work from your own GitHub account. If one person uploads everyone's text from a shared document, the history shows one author and the other members have no verifiable evidence. A shared drafting document is fine; the final text of each section is committed by its owner.
Cite only what exists. Every commit, pull request, branch and file named in this table is checked. Citing evidence that is not in the repository is an academic-honesty matter, not a formatting slip.
Every member appears with a distinguishable contribution. Two rows with identical wording are marked as vague claims. If a member did not contribute, write that plainly in their row rather than leaving it blank or covering for them.

0.2 Changes since the P1 baseline

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Change ID	What changed (IDs affected)	Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on

CH-01	UC-01 rewritten turn by turn in 4.2 with step (0), alternate flows and an acknowledgment	P1 feedback C3: main flow was a single line, alternate flow one sentence	5.2.1 sequence diagram is built from it	Team meeting , Nov 4
CH-02	Second and third expanded use cases added: UC-02, UC-03 (4.2)	P1 feedback C3: only one use case was written out in full	5.2.2 sequence diagram is built from UC-02; 9.1 traces FR-10 to it	Team meeting , Nov 4
CH-03	Section 3.5 matrix replaced by a pointer to 9.1; every requirement now maps to a use case ID (FR-01 to FR-12, QR-01 to QR-06, CON-01 to CON-04)	P1 feedback C3: matrix pointed to phrases, and UC-01 claimed all twelve FRs	9.1 extends the chain to classes, components and ADRs	Team meeting , Nov 4
CH-04	Acceptance tests of FR-02 and FR-04 rewritten with a number and a pass condition; QR-02 reworded with a measurable threshold in the same form as QR-01; IDs unchanged	P1 feedback C2: not every requirement had a concrete acceptance test	QAS-01 in 6.1 cites the revised QR	Team meeting , Nov 4
CH-05	Section 2.2 adds answers from UTD students on how they find events today; 2.6 adds the S-01 scrape result	P1 feedback C1: need and feasibility were assumptions	Feasibility verdict now rests on measured evidence	Team meeting , Nov 4
CH-06	Repository given a structure (/docs/p2, /docs/p2/diagrams, /spikes); every section committed by its owner through a reviewed pull request	P1 feedback C4/C5: repository held only the README, so contributions could not be verified	0.1 cites commit hashes and PR numbers	Team meeting , Nov 4
CH-07	Roles in 10.1 stated as responsibilities instead of section numbers	P1 feedback C4	None	Team meeting , Nov 4
CH-08	Every milestone in 10.2 has one named owner	P1 feedback C4: five of seven milestones were owned by "Team 5 Members"	None	Team meeting , Nov 4

CH-09	Risk R5 filled; R1 to R4 updated; design risks added (10.3)	P1 feedback C4: R5 was blank	New risks point to S-01 and ADR-02	Team meeting , Nov 4
CH-10	AI-use declaration completed by every member (10.5)	P1 feedback C4: declaration was empty	Commits with an AI co-author line are matched in 10.5	Team meeting , Nov 4
CH-11	All template guidance, example and warning boxes removed before export	P1 feedback C4: guidance boxes were left in	None	Team meeting , Nov 4
CH-12	Open questions Q1 to Q3 closed with their answers (11)	Answered by spike S-01 and ADR-02	Assumptions in 2.5 confirmed or revised	Team meeting , Nov 4

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Start with your P1 feedback. Every point raised in your P1 comment should appear here as a change, or with one sentence explaining why you kept the original. This table is where the baseline-and-change-control idea from Chapter 5 becomes real.
Requirement IDs are never reused. A removed requirement stays in section 3 marked "Withdrawn (CH-nn)" so the traceability matrix does not silently lose it.
Budget: half a page.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
CH-02 · QR-02 rewritten as quality-attribute scenario QAS-01 · Why: P1 feedback said "fast enough" had no response measure · Knock-on: ADR-02 now cites QAS-01 · Agreed 30 Oct.

1 to 3 Introduction, Overall Description and Requirements (carried forward)

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (P1 sections 1–3 pasted below, then fixed) (delete this line before submitting)

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Paste your P1 sections 1, 2 and 3 here and bring them up to date. Keep their numbering and IDs.
Fix before you build on it: any requirement that is still untestable, any source that still says "pending", any open question your P1 feedback flagged. Analysis and design inherit every ambiguity you leave in section 3.
Use cases: section 3.4 keeps the use-case diagram and the list at actor-goal level. The detailed expansion now lives in section 4; do not copy it twice.
Traceability: replace the P1 matrix in section 3.5 with a one-line pointer to section 9.1, where it is extended.
If nothing in a subsection changed, keep it and say so in one line under its heading ("Unchanged since v0.x").

1 Introduction

1.1 Purpose of this document and of the system

Whoosh Events is a website that gathers UTD's events (clubs, social, community) into one list, and shows students where to submit their own events on the official source sites. Currently, information about UTD Events is scattered across multiple websites, so our goal is to consolidate that information into one website.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Scraper that finds all events from multiple sources	No comments over event details
Loaded list[block style] of events with filter	Direct event creation
RSVP Option/ add to personal calendar	Calendar view of events
A page that teaches you how to create a UTD event (club, community, social)	Log in page
Event card- card template that fits the information relevant to the specific event	Event Recommendation

Why the exclusions were excluded:
Direct event creation - To avoid events that were planned without permission, we decided to avoid having direct event registration
No comments over event details - This should fall on the Event Organizer to make their contact info very clear for any questions attendees may have.
Calendar view of events - Calendar view can make browsing difficult; list view will make it easier for users to find events they are interested in.

No log in page - not required to see public events and no security is needed to look at the number of events as a whole.
No event recommendation - Users should be able to know what they like or dislike. The goal of our website is to consolidate information for easier access, not to make recommendations.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
Event	A scheduled UTD activity (club, social , or community) listed on a source site in 1.4
Event Card	A fixed layout block that shows one event details in the event list
Event List	The scrollable list of event cards on the main page
Tag	A category or label on an event (for example “social” or “career”) that users can filter by. Our Label list is Social, Academic, Career, Arts & Culture, Sports & Recreation, Service & Volunteering, Cultural.
Scraper	The component that automatically reads event pages from the source sites and extracts event data.
Event organizer	the club, department, or person who posts an event on a source site.
Source site	An external site the scraper reads: the UTD calendar, or the Nebula labs club Calendar.
RSVP	A user’s signal that they plan to attend an event

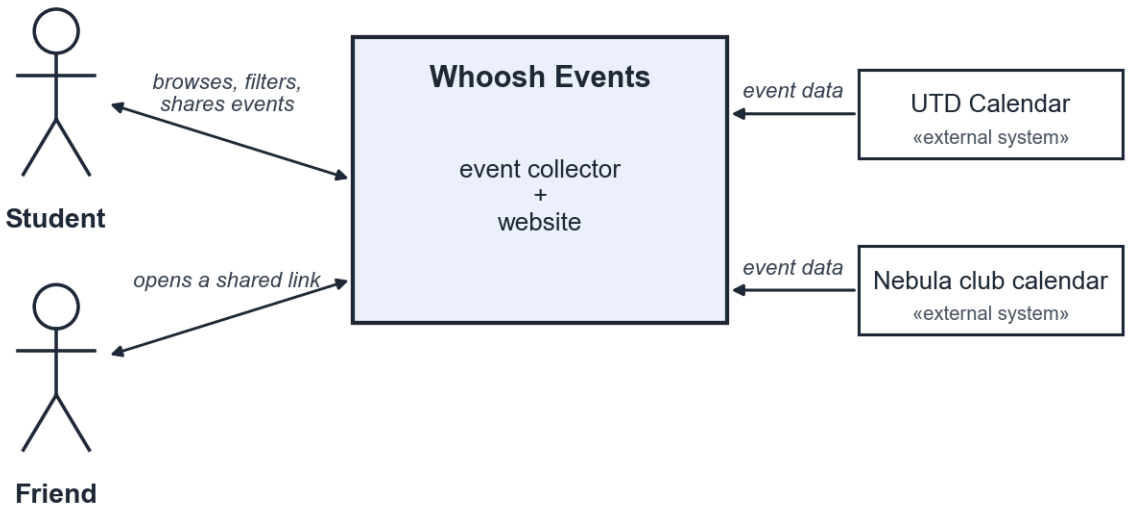
1.4 References

[R1] UTD Calendar - https://calendar.utdallas.edu/calendar
[R2] Nebula Labs Club Calendar - https://clubs.utdnebula.com/events?page=1

2 Overall Description

2.1 Product perspective and context

Figure 2.1 Context diagram



The boundary surrounds Whoosh Events itself: the event collector and the website, which are the only parts we build and maintain. Students and their friends use the website from outside the boundary. The UTD Calendar and the Nebula club calendar are external systems we only read from, under their robots.txt rules (CON-03); we did not build them and cannot change them.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Site Users	Start a use case	One place that shows events from all UTD groups, including ones they do not already follow, early enough to attend, and filters to find what interests them	Poll of 7 UTD students on Nov 5, 2026 (How do you find out about UTD events?). Each relies on one channel: Instagram, a club Discord, a WhatsApp free-food group, walking past the Student Union, MSA announcements, the

			UTD Calendar or university emails; the official channels carry only academic and university events, and none mentioned Nebula. One hears of events only after they are over. So students see only the groups they already follow, and too late.
Developer Team	Builds and Maintains the Site	A scraper to grab information from UTD pages, and a website to display the information obtained in a formal layout.	This was decided on during the 10/14/26 Team 05 meeting. These are fundamental parts of our idea.
Event Organizers(clubs, departments)	Indirect user, content provider	Their events shown accurately, and a clear way to learn how to host one	From the source sites in 1.4: every event names its organizer, and each site has its own process for posting an event. FR-04 points organizers to that process instead of replacing it.
Nebula Labs and UTD Calendar Maintenance teams	Operate the source sites, never use our system	Scraping that doesn't overload or break their sites	From the sites' robots.txt files, read in Nov 2026: the UTD Calendar asks automated readers to wait 1 second between requests, and Nebula allows all pages. CON-03 follows these rules.

Course instructor & Teaching Assistants	Evaluate the project	Requirements that are testable and within scope	stated in the p1 template

2.3 User characteristics

Users are assumed to have basic knowledge to navigate the internet. They should already know that the site is for navigating a list of UTD events. They cannot be assumed to know what events are available. It may also be assumed that they are accessing the site from a laptop, desktop, or mobile device. Layouts must work on a phone screen, and the site should be readable for users with accessibility needs, for example keyboard navigation and sufficient text contrast.

2.4 Constraints

Event data comes only from the sources, which are the source sites (Nebula labs UTD clubs & UTD Events calendar). Users cannot create events directly on the site. P2 must be submitted by November 8, 2026. Work is managed in the team Github repo, with testing in preproduction before merging to production.

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Every event on the source sites in 1.4 has a readable title, date and start time.	An event with no date cannot be placed in the list; fields that are only missing show Not listed (FR-07).
A2	Both source sites stay publicly accessible without login and don't block automated access.	The scraper returns no events, and the event list is empty or stale.
A3	The same event is not posted on both sources, or can be matched when it is.	The same event appears twice in the event list, which breaks FR-03.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible, provisional	Both source sites are public event pages (Nebula labs UTD clubs & UTD Events Calendar). The UTD Calendar also offers a public events feed, and both robots.txt files allow reading (2.2). Spike S-01 (section 8) measures a real scrape of both sites.
Operational	Feasible	The site is read-only and has no accounts, so there is little to operate once deployed.

Schedule	Feasible with risk	6 members, and the plan in 10.2 runs to the P3 deadline on December 9.
Economic / resource	Feasible	No budget is needed. It depends on the website being hosted for free and student time.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-01	The system shall have at least 20 events listed on one page.	MUST	1.2	A tester can confirm this by loading our website and counting the number of events listed on the page.
FR-02	The system shall have filters to display events by a specific “tag” from the list defined at 1.3	MUST	1.2 & 1.3	A tester picks 5 different tags from the list in 1.3 and filters by each one. Pass if, for all 5 tags, every event shown has the chosen tag and no upcoming event with that tag on the source sites in 1.4 is missing.
FR-03	An event that appears on more than one source site from our 1.4 list shall appear only once in the event list.	SHOULD	1.4, 2.2	A tester finds 3 events listed on at least two of sources listed on at 1.4. Pass if each appears exactly once in the event list.
FR-04	The system shall have a 'How can I host an event at UTD?' page that links to the official page for submitting an event on each source site in 1.4.	COULD	1.2	A tester opens the page and follows every link on it. Pass if the page has one link for each of the 2 source sites in 1.4 and each link opens that site's official page for submitting an event.

FR-05	The system shall show every event from source 1.4 in our event list.	MUST	1.2	A tester picks 10 upcoming events from each source of two sources from our sources list. Pass if all selected events appear in our event list.
FR-06	The system shall have an “Add to Calendar” feature that allows users to download .ics files.	COULD	1.2	A tester can confirm this by accessing any event on our website and clicking on the “Add to Calendar” icon. Then, confirm that the .ics file is downloaded.
FR-07	Each event card shall show the event's title, date, start time, location, organizer, tags and a link to the source page.	SHOULD	1.2	A tester checks 10 event cards, at least 3 from each source site. Pass if every card shows all seven fields, or 'Not listed' when the source site leaves a field blank.
FR-08	The system shall let a Site User filter the event list by event organizer.	SHOULD	2.2	A tester filters by 3 different organizers. Pass if every event shown each time belongs to that organizer.
FR-09	The system shall show an event’s full description when a Site User selects its event card.	SHOULD	2.2	A tester selects 5 event cards. Pass if each shows the same description as on its source site.
FR-10	Each event card shall have a “Copy link” button that copies a URL opening that event on our site.	COULD	2.2	A tester copies the link from 5 events and opens each in a new browser window. Pass if each opens the same event.
FR-11	The system shall load the next 20 events when a Site User reaches the end of the event list.	SHOULD	2.2	With more than 40 upcoming events, a tester scrolls to the end of the list. Pass if 20 more events appear.

FR-12	Each event card shall show the name of the source site the event came from.	COULD	2.2	A tester checks 10 event cards, at least 3 from each source site. Pass if each shows the correct source site name.

3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	The event list page shall finish loading within 3 seconds on at least 19 of 20 page loads over a connection of 10 Mbps download with 50 ms latency.	SHOULD	2.2	A tester creates a custom Chrome DevTools network profile (10 Mbps download, 50 ms latency), loads the event list page 20 times and records each load time. Pass if at least 19 loads finish in 3 seconds or less.
QR-02	After a student chooses a tag, the filtered event list shall appear within 2 seconds on at least 19 of 20 tries over a connection of 10 Mbps download with 50 ms latency.	SHOULD	1.2	A tester uses the same Chrome DevTools network profile as QR-01, chooses a tag 20 times (rotating through 5 tags) and records the time from choosing the tag to the list appearing. Pass if at least 19 of the 20 finish in 2 seconds or less.
QR-03	An event posted on a source site shall appear in the event list within 2 hours for at least 9 of 10 newly posted events.	MUST	2.2, 5	A tester records when 10 new events are posted on the source sites in 1.4 during one week, then checks when each first appears in our event list. Pass if at least 9 appear within 2 hours.

QR-04	At least 19 of 20 randomly sampled events shall show the same title, date, time and location as on their source site.	MUST	2.2	A tester picks 20 events at random from our event list and compares the four fields with the source site. Pass if at least 19 matches on all four.
QR-05	Every page shall display without horizontal scrolling on a screen 375 pixels wide.	SHOULD	2.3	A tester opens every page in Chrome DevTools device mode at 375 pixels wide. Pass if no page needs horizontal scrolling.
QR-06	When a source site cannot be reached during a refresh, the event list shall keep all of that site's events from the last successful refresh.	SHOULD	2.5	A tester blocks one source site's address and runs a refresh. Pass if every event from that site that was listed before the refresh is still listed after it.

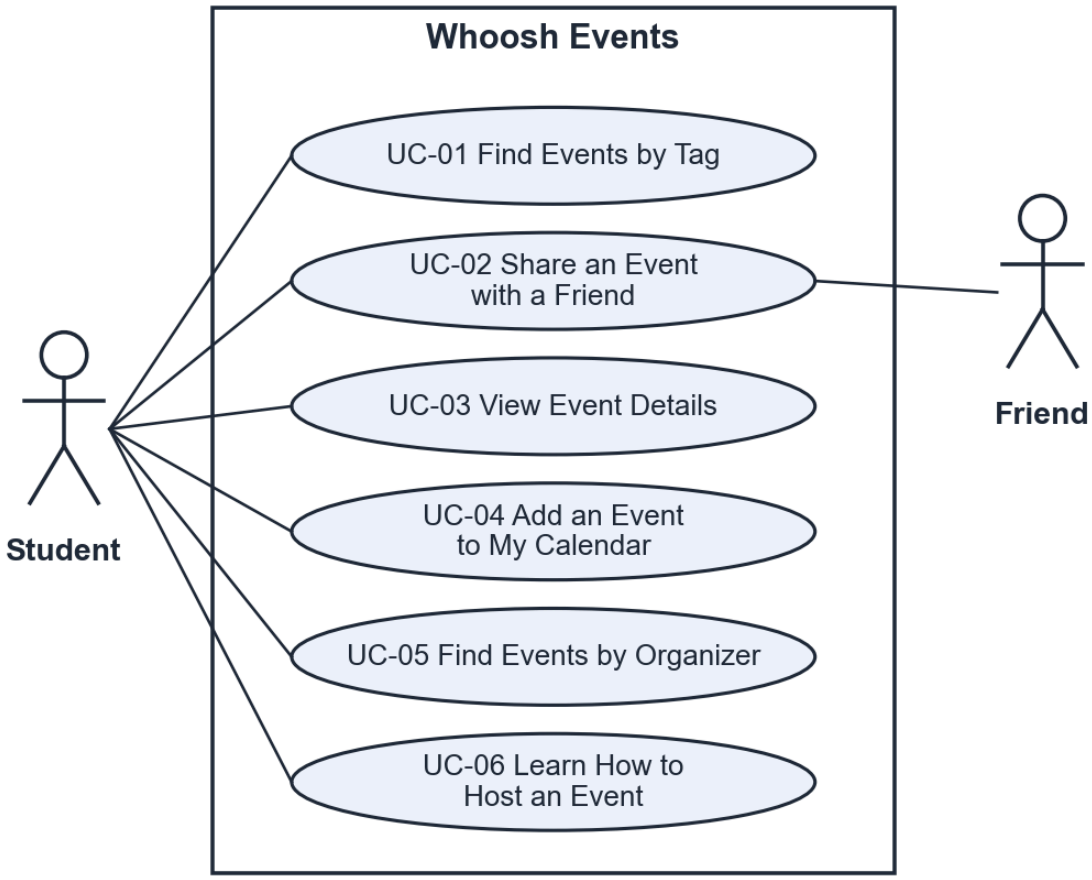
3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall not allow Site Users to create their own events.	2.4, 1.2	UTD events must be approved through UTD's own process before they are posted. Our site has no way to check that approval, so allowing event creation would let unapproved events appear as if they were official.
CON-02	The system shall get event data only from the source sites listed in 1.4.	2.4	The UTD Calendar and the Nebula Labs Club Calendar are UTD's official places for Uposting events. Taking events from any other site could show unapproved or fake events.
CON-03	The scraper shall read only the pages that each source site's robots.txt file allows.	2.2, 2.5	The owners of the source sites set these rules, not us. Ignoring them could get our scraper blocked, which would break

			assumption A2 and leave the event list empty.
CON-04	The system shall run on a free hosting service with no paid costs.	2.6	The team has no budget for this project, as stated in the economic feasibility verdict in 2.6.

3.4 Initial use cases

Figure 3.4 Use case diagram (actor-goal level)
m



Redrawn since P1 with the use case IDs used in sections 4, 5 and 9. The P1 diagram showed the scraper and the event organizers; the scraper is background work no actor sees (see 4.3), and organizers post on the source sites, not on our system. The three use cases marked “expanded” are written out turn by turn in 4.2.

UC-01 Find Events by Tag. Student finds upcoming events on a topic. FR-01, FR-02, FR-05, FR-07, FR-11. Expanded.

UC-02 Share an Event with a Friend. Student sends one event to a friend, who opens it. FR-07, FR-09, FR-10, FR-12. Expanded.

UC-03 View Event Details. Student sees everything about one event. FR-03, FR-07, FR-09, FR-12. Expanded.
UC-04 Add an Event to My Calendar. Student keeps an event in their own calendar. FR-06.
UC-05 Find Events by Organizer. Student sees what one club or office is running. FR-08.
UC-06 Learn How to Host an Event. Student finds where to submit their own event on each source site. FR-04.

3.5 Requirements to use case traceability

The P1 matrix is replaced by the full traceability table in section 9.1, which maps every requirement to a use case ID and on to classes, components and decisions.

4 Use-Case Analysis

Chapter 7, first half. Here you take the use cases from section 3.4 and write the most important of them out turn by turn, still with the technology switched off.

4.1 Use cases selected for analysis

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Use case	Name	Why it was selected	Non-trivial steps
UC-01	Find Events by Tag	Carries most of the product: the main reason students visit	Steps 0, 2, 4
UC-03	View Event Details	Touches the most event fields (FR-03, FR-07, FR-09, FR-12, QR-04), including events merged from both source sites	Step 2
UC-02	Share an Event with a Friend	Two people and a link that can go out of date: the event can end, be removed or change before the friend opens it; tests QR-03 and QR-04	Steps 2, 4

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Analyze at least three use cases, including the two you specified in full at P1. We recommend that one of them involves an external system or a failure that the actor can see. Choose by risk and value, not by ease: the use case that carries most of the product, the one that touches most of your quality requirements, the one you are least sure you can build. Budget: a quarter of a page.

4.2 Expanded use cases

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)
Copy the block below once for each use case in 4.1.

UC-01 Find Events by Tag

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Primary actor	Student
Supporting actors and external systems	None during this use case (events were collected from UTD Calendar and Nebula beforehand)
TUCBW (this use case begins when)	The student opens Whoosh Events to look for a certain kind of event
TUCEW (this use case ends when)	The student sees the upcoming events that have the tag they chose
Preconditions	The student has a web browser and an internet connection
Postconditions	The student sees the filtered list; nothing in the system has changed
Traces to requirements	FR-01, FR-02, FR-05, FR-07, FR-11, QR-01, QR-02, QR-03, QR-06

Actor input / action	**System response**
	(0) The system shows the first 20 upcoming events, soonest first. Each shows its title, date, start time, location, organizer and tags. It also shows all available tags and the time the list was last updated.
(1) The student chooses a tag (for example “Career”).	(2) The system shows only the upcoming events that have that tag, soonest first,

	and how many match (for example “14 events”).
(3) The student reaches the end of the events shown.	(4) The system adds the next 20 matching events, or says “No more events” when all are shown.

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(1)	The student chooses more than one tag	The system shows the events that have any of the chosen tags	Rejoins (2)
A2	(2)	No upcoming event has the chosen tag	The system says “No upcoming events match this tag” and offers to clear the filter	Ends
A3	(0)	The last update is more than 2 hours old (a source site could not be reached)	The system shows the events from the last successful update, with “Last updated at” and the time	Rejoins (1)
A4	(2)	The student clears the chosen tags	The system shows all upcoming events again	Rejoins (0)

UC-03 View Event Details

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Primary actor	Student
Supporting actors and external systems	The event’s source site (UTD Calendar or Nebula), when the student opens the original page

TUCBW (this use case begins when)	The student sees an event in the list they want to know more about
TUCEW (this use case ends when)	The student sees the event's full details, or its original page
Preconditions	The student has a web browser and an internet connection
Postconditions	The student has seen the event's details; nothing in the system has changed
Traces to requirements	FR-03, FR-07, FR-09, FR-12, QR-04, QR-05

Actor input / action	**System response**
	(0) Shows the event list. Each event shows its title, date, start time, location, organizer, tags and the name of its source site
(1) Selects an event	(2) Shows the event's full description together with all of the fields above, and a link to the event's original page
(3) Asks to see the original page	(4) Opens the event's page on its source site

Alternate flow	**At step**	**Condition**	**What happens**	**Ends or rejoins at**
A1	(2)	A field (for example the location) is blank on the source site	Shows "Not listed" for that field	Rejoins at (3)
A2	(2)	The event is listed on both source	Names both source sites and gives a link to each	Rejoins at (3)

		sites, so it was merged into one entry		
A3	(2)	The event ended while the student was looking at the list	Shows the details marked “This event has ended”	Ends

UC-02 Share an Event with a Friend

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Primary actor	Student
Supporting actors and external systems	The friend who receives the link
TUCBW (this use case begins when)	The student finds an event they want to tell a friend about
TUCEW (this use case ends when)	The friend sees that event’s details on Whoosh Events
Preconditions	The student and the friend each have a web browser and an internet connection
Postconditions	The friend sees the current details of the shared event; nothing in the system has changed
Traces to requirements	FR-07, FR-09, FR-10, FR-12, QR-03, QR-04

Actor input / action	**System response**
	(0) The system shows the event list; each event offers “Copy link”.

(1) The student asks to copy an event's link.	(2) The system copies a link that opens that event and says "Link copied".
(3) The friend opens the link (the student sends it to them outside our site).	(4) The system shows that event's full details: title, date, time, location, organizer, tags, description, source site and a link to its original page.

Alternate flow	**At step**	**Condition**	**What happens**	**Ends or rejoins at**
A1	(2)	The browser does not allow copying	The system shows the link so the student can copy it by hand	Rejoins (3)
A2	(4)	The event has already ended	The system shows the details marked "This event has ended", plus upcoming events with the same tags	Ends
A3	(4)	The event was removed from its source site after the link was shared	The system says "This event is no longer listed" and shows upcoming events from the same organizer	Ends
A4	(4)	The event's time or place changed after the link was shared	The system shows the updated details marked "Updated since this link was shared"	Ends

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
The rules from Lesson 13, all of which are checked:
Step (0) is what the actor sees before touching anything. It is often skipped, and it is where many missing requirements are found.
One turn per row. An actor action on the left, the system response on the right. Two steps merged into one row become a missing message on your sequence diagram in section 5.
End with an acknowledgment the actor can perceive. No acknowledgment, no TUCEW, no use case.
No background processing. If the actor neither sees it nor waits for it, it belongs to another use case.
Precondition or alternate flow? If it can be false while the system works normally, it is a flow you must handle, not an assumption.
No technology yet. No screen widgets, no database, no framework names. "Taps the green button" and "queries the PostgreSQL table" both fail the analysis check.
Every item the actor column names should already be an attribute in your domain model (5.1). Where it is not, you have found one.
Budget: about one page per use case.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
UC-02 Find an Open Space · Primary actor: Driver · External system: Lot Sensor Network · TUCBW the driver asks for parking near a destination building · TUCEW the driver sees the recommended lot with its number of open spaces.
(0) The system shows the campus lots with the time of the latest occupancy update. (1) The driver chooses a destination building and permit type. (2) The system shows the three nearest lots the permit allows, each with open spaces and walking time. (3) The driver selects a lot. (4) The system shows the lot entrance and confirms the count is no older than 60 seconds.
A1 at step (2): no permitted lot has an open space · the system shows the nearest lot with expected turnover and the visitor option · ends.

4.3 Self-review log

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-01	Q2, Q3, Q7 and Q8: step (0), one turn per row, a testable postcondition,	The P1 version had no step (0), the whole flow in one line, an untestable postcondition (they	Rewritten as steps (0) to (4), one turn per row, with a testable postcondition; A1 to

	alternates named by their step	close the website) and an alternate with no step	A4 each name the step they branch from
UC-01	Q1 and Q6: are actors roles that take part, and are preconditions true assumptions?	P1 listed Organizations and Clubs as actors, though they never act, and assumed events exist on the source sites, which can be false	Actors removed; the precondition became alternates A2 (no matching event) and A3 (a source site could not be reached)
UC-02	Q4: is every item the actor supplies or sees an attribute in the domain model?	Alternate A4 needs the time the link was shared, and no class held it	Added the class SharedLink (shared at) to the domain model in 5.1
UC-03	Q4 (same check)	Alternate A2 needs one original page address per source site for a merged event, but Event had room for only one	Added the class SourceListing (original page address), one or two per Event
Refresh Event Data (first draft of 4.1)	Q9: is there background processing that should not be here?	It runs on a timer; no actor starts it, sees it or waits for it	Not expanded as a use case; freshness stays a quality requirement (QR-03) and shows up to the student as alternate A3 of UC-01
All three	Q5 and Q10: does each end with something the actor can see, and does any row name a technology, a store or a screen widget?	None found	No change needed

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Run the review checklist from Lesson 13 on each expanded use case before anyone draws a diagram, and record what each check found, including any defects you fixed. Report honestly; a clean result is acceptable when it is true. Budget: a quarter of a page.

5 Analysis Models

Chapter 7, second half. What has to exist inside the system for the use cases in section 4 to happen, and how those parts talk to each other.

5.1 Domain model

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

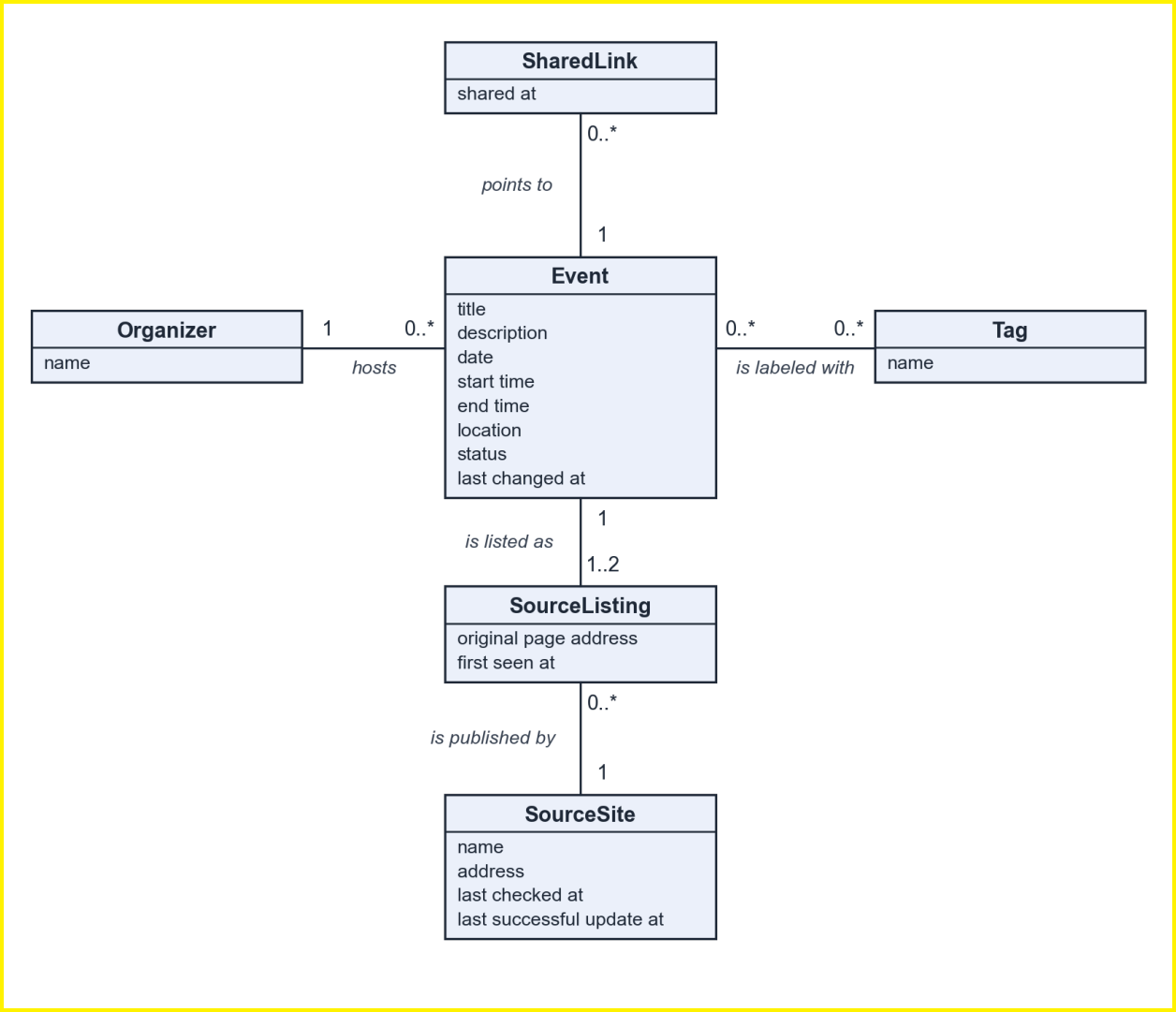


Figure 5.1 Domain model (conceptual classes, attributes, associations with multiplicities)

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
Event	One happening a student can attend, as Whoosh Events lists it (merged into one entry when both source sites list it)	Title, description, date, start time, end time, location, status (upcoming, ended, removed), last changed at	Hosted by 1 Organizer; labeled with 0..* Tag; listed as 1..2 SourceListing; pointed to by 0..* SharedLink
Tag	A topic label a student can filter by, such as Career or Social	name	Labels 0..* Event
Organizer	The club, department or office that runs an event	name	Hosts 0..* Event
SourceListing	The record of an event as one source site publishes it	Original page address, first seen at	Belongs to 1 Event; published by 1 SourceSite
SourceSite	An official UTD site we collect events from (UTD Calendar, Nebula club calendar)	Name, address, last checked at, last successful update at	Publishes 0..* SourceListing
SharedLink	A link a student copied to send one event to a friend	Shared at	Points to 1 Event

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Conceptual classes only: things and records from the problem domain, with attributes and associations. No operations, no data types, no database tables, no screen classes.
Every item the actor column of section 4.2 mentions is an attribute somewhere in this model.
Every association has a multiplicity at both ends.
Budget: one page including the diagram.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING

Lot (name, zone, capacity) 1 ── 0..* Space (number, type) ; Space 1 ── 0..* OccupancyReading (time, state) ; Permit (type, valid until) * ── 1..* Lot "is valid in" ; Building (name) * ── * Lot "is near" (walking time).

5.2 Object interaction models

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

5.2.1 UC-01 Find Events by Tag

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

(a) Non-trivial steps.

Non-trivial: step (0), because the system must fetch the upcoming events, the tags in use and the time of the last update; step (2), because it must find the events with the chosen tag, order them and count them; step (4), because it must fetch the next 20 matches or decide there are none. Steps (1) and (3) are the student's own actions; each appears below only as the message that starts the next step.

Objects: EventBrowser is the interaction object that runs the student's visit and is in control throughout. EventCatalog is the interaction object that holds the collected Events and their Tags (5.1). SourceSite is the domain class from 5.1.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
0a	EventBrowser	Asks for the first 20 upcoming events, soonest first	Today's date, the page size of 20	EventCatalog
0b	EventBrowser	Asks for the tags in use	none	EventCatalog
0c	EventBrowser	Asks when each site was last updated successfully	none	SourceSite
0d	EventBrowser	Shows the events, the tags and the last-updated time	The event list, the tag list, the last update time	Student
1	Student	Chooses a tag	The tag name	EventBrowser

2a	EventBrowser	Asks for the upcoming events with that tag, soonest first	The tag name	EventCatalog
2b	EventBrowser	Counts the matching events	The matching events	EventBrowser (itself)
2c	EventBrowser	Shows the matching events and how many match	The matching events, the count	Student
3	Student	Reaches the end of the events shown	How many events are shown	EventBrowser
4a	EventBrowser	Asks for the next 20 matching events	The tag name, how many are already shown	EventCatalog
4b	EventBrowser	Adds the next 20 matching events	The next events	Student

(c) Analysis sequence diagram.

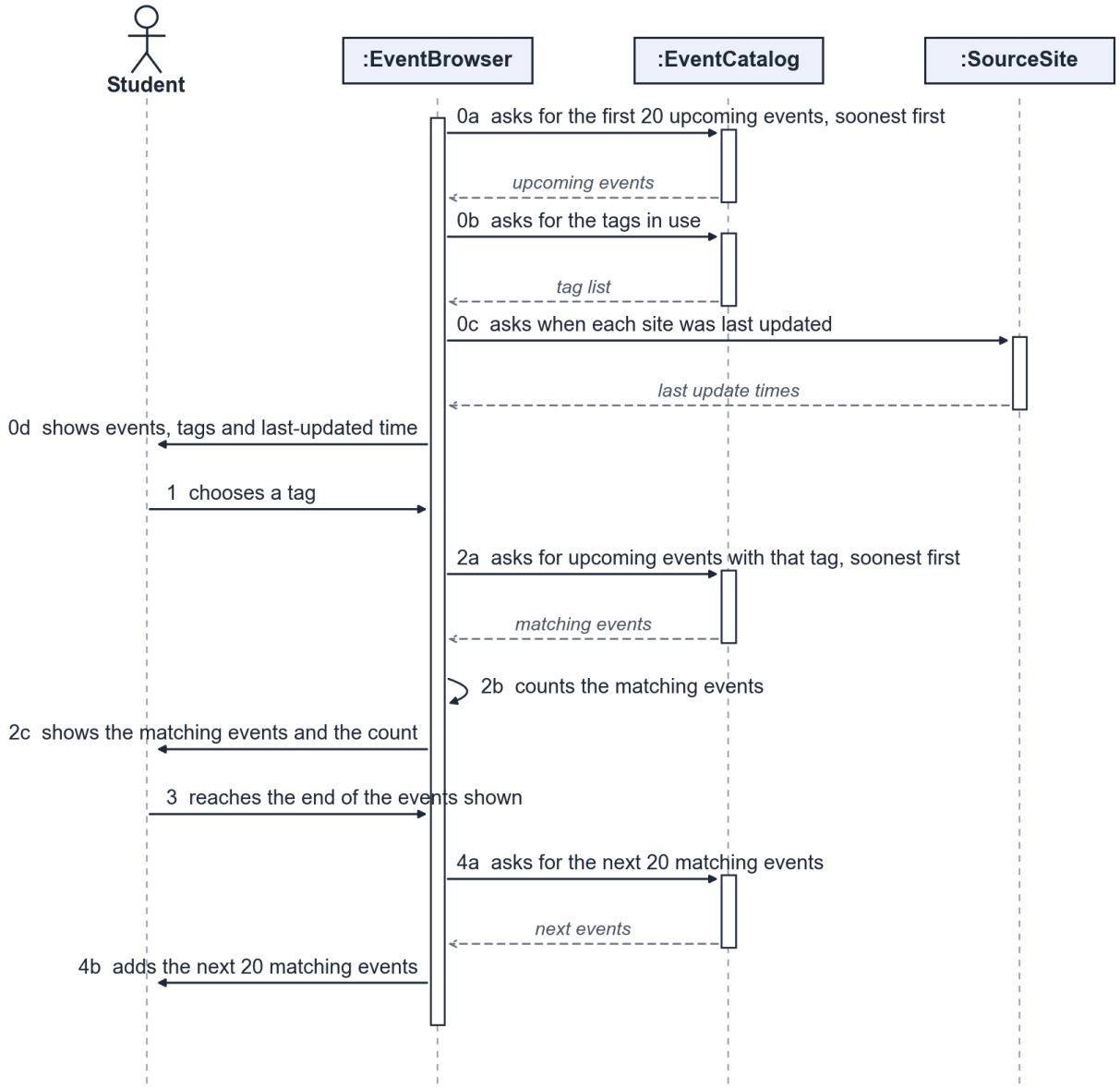


Figure 5.2.1 Analysis sequence diagram for UC-01; message numbers match the scenario table

5.2.2 UC-02 Share an Event with a Friend

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

(a) Non-trivial steps.

We chose UC-02 because it is our riskiest use case: alternates A2 to A4 depend on how fresh our copy of the source sites is (QR-03, QR-04). Non-trivial: step (2), because the system must make a link that identifies the event and record when it was shared; step (4), because it must find the event the link points to, check that it is still upcoming and whether it changed since it was shared, and only then show it. Trivial: step (0) is the same event list as UC-01 step (0),

already modelled in 5.2.1. Steps (1) and (3) are the student's and the friend's own actions; each appears below only as the message that starts the next step.
Objects: EventBrowser (from 5.2.1) is in control; here it serves the student and, when the link is opened, the friend. LinkRegistry is the interaction object that keeps each SharedLink (5.1). EventCatalog holds the collected Events (5.1).

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	Student	Asks to copy an event's link	The chosen event	EventBrowser
2a	EventBrowser	Asks for a link to the event	The chosen event	LinkRegistry
2b	LinkRegistry	Records the link with the time it was shared	The event, the current time	LinkRegistry (itself)
2c	EventBrowser	Copies the link and says "Link copied"	The link	Student
3	Friend	Opens the link	The link	EventBrowser
4a	EventBrowser	Asks which event the link points to and when it was shared	The link	LinkRegistry
4b	EventBrowser	Asks for the event's current details and status	The event	EventCatalog
4c	EventBrowser	Checks the event is still upcoming and unchanged since it was shared	Status, last changed at, shared at	EventBrowser (itself)
4d	EventBrowser	Shows the full details and the link to the original page	Title, date, time, location, organizer, tags, description, source site, original page address	Friend

(c) Analysis sequence diagram.

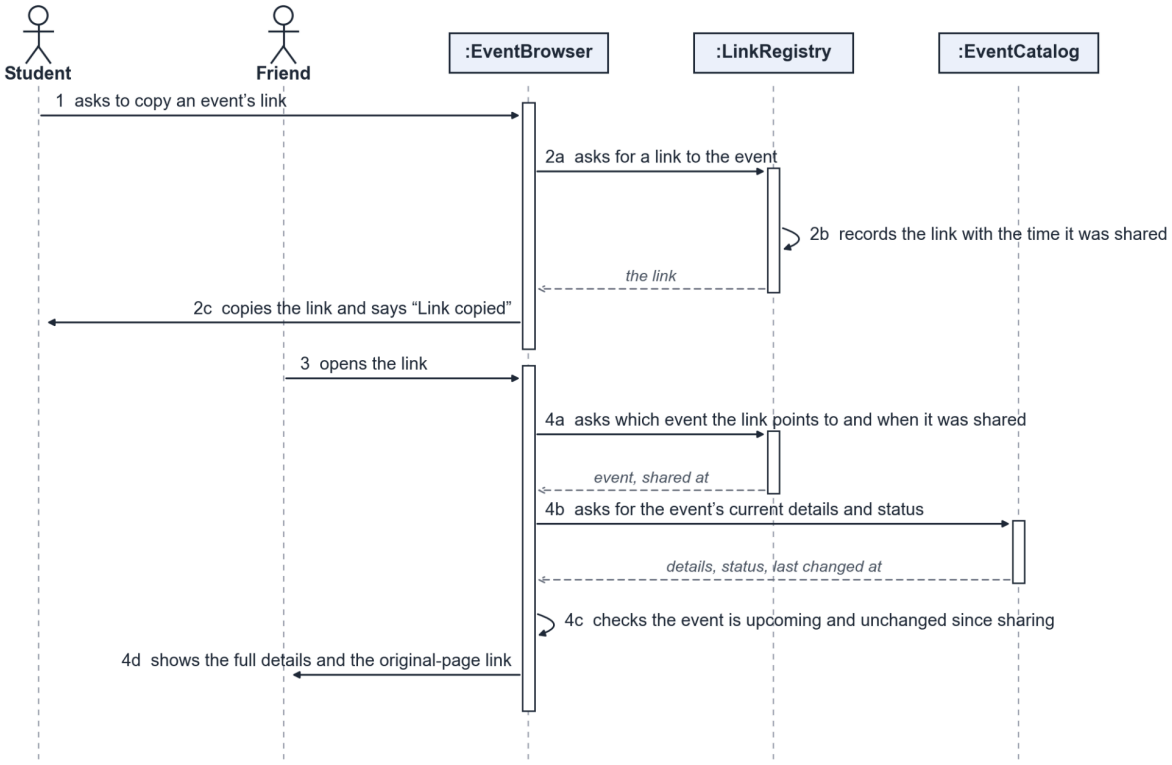


Figure 5.2.2 Analysis sequence diagram for UC-02; message numbers match the scenario table

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Write each step as a declarative sentence first, subject, verb, object, then fill the columns. Tables filled directly produce rows with no subject.
One message per table row, same numbers. Use the four translation cases from Lesson 14; the diagram is a conversion of the table, not a new drawing.
Messages are verb phrases, not operation signatures. "asks for the routes calling at the stop" is analysis; "getRoutes(stopId: int): List<Route>" is Chapter 9 and is not credited here.
Every object on the diagram is a class in 5.1 or an interaction object you name and explain in one line. One object is clearly in control of the interaction.
Budget: one to one and a half pages per use case.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
Step 2 · Subject: LotFinder · Action: asks for the lots the permit allows · Data: permit type, destination · Object acted upon: PermitRules.

5.3 Consistency cross-check

► OWNER: Atharva · due Sat Nov 7, 6 PM (delete this line before submitting)

Check	Result (pass, or what failed)	Fixed in
K1 Every item in an actor column (4.2) is an attribute in the domain model (5.1)	pass	
K2 Every object on a sequence diagram is a domain class or a declared interaction object	pass	
K3 Every message on a sequence diagram corresponds to exactly one scenario-table row	pass	
K4 Every non-trivial step in 5.2 (a) has at least one scenario-table row	pass	
K5 Every analyzed use case traces to at least one functional requirement	pass	
K6 The last messages of each diagram make the use case postcondition true	Fail (the postcondition for UC-01 says the student sees the filtered list, but figure 5.2.1 displays the last message 4b, which says that the website “adds the next 20 matching events”)	Fixed in 4.2, changed the wording of the postcondition to match the 5.2.1 diagram

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Run the six checks across the work of different authors; they are designed to catch the seams between people. A failed check that you found and fixed is more convincing than a column of passes. Budget: a quarter of a page.

6 High-Level Design (Architecture)

Chapter 8. The rule now reverses: a high-level design that does not say where things run, what they are built from and what that choice cost has not been done.

6.1 Architectural drivers: quality-attribute scenarios

QAS	Quality attribute	Stimulus	Artifact	Response	Response measure	From
QAS-01	Performance	A student opens the event list, or applies a tag filter, on a 10MBPS connection with a 50 ms latency.	C3 website on Vercel	Shows the irst 20 event cards, or the filtered list.	Page Finished loading within 3 seconds on atleadt 19 out of 20 loads	QR-01,QR-02
QAS-02	Timeliness	An organizer posts a new event on the UTD Calendar or nebula site	C1 Event collector and the stored events.json	The event is collected, saved and shown on the site.	Within 2 hours for at least 9 of 10 new events	QR-03
QAS-03	Availability	One Source site cannot be reached during a scheduled refresh	C1 Event Collector and C2 Event Storage	Events from that site's last good refresh stay in events.json and stay on the site	100% of that site's previously listed events still shown after the failed refresh	QR-06

Constraints shaping the architecture: CON-01 (users cannot create events, so the site is read-only), CON-02 (only the two source sites), CON-03 (robots.txt rules), CON-04 (free hosting only).

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
QAS-01 · Performance · Stimulus: 400 drivers request lot status during the 7:45 to 8:15 AM peak · Artifact: the occupancy service · Response: returns current open-space counts · Measure: within 2 s for 95% of requests · From QR-02.

6.2 Architectural style and alternatives

► OWNER: Habib · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Style considered	Drivers it serves	What it costs	Verdict
Repository (shared data file) with a scheduled batch collector; browsers talk to a static site (client-server)	QAS-01, QAS-02, QAS-03, CON-04	Events can be up to one refresh old. The site redeploys on each refresh. The collector must remove duplicates (or not include them)	Chosen
Client-server with live fetch: the site calls both source sites on every page view	QAS-02 Always current	Page speed depends on two outside sites (breaks QAS-03). Needs an always -on server (CON-04)	Rejected
Event-driven / microservices: separate services connected by a message queue	Scaling and easy change	The source sites send no push events, so there is nothing to react to. Several hosted services will not fit a free plan (CON-04). Too much for a 6-person team to build by December.	Rejected

We chose the repository style because the 3 scenarios all depend on keeping a saved copy of events. The collector (C1) writes events.json on a schedule and the website (C3) only reads it, so the page speed in QAS-01 does not depend on the source sites. A refresh within an hour or two meets requirement QAS-02. A failed refresh leaves the saved file unchanged, which meets QAS-03. Live fetching has no saved copy and cannot meet QAS-01 or QAS-03.

Event-driven or microservice designs add parts amd cost that CON-04 and our team size don’t allow.

6.3 Component view

► OWNER: Austin · due Fri Nov 6, 11:59 PM (delete this line before submitting)

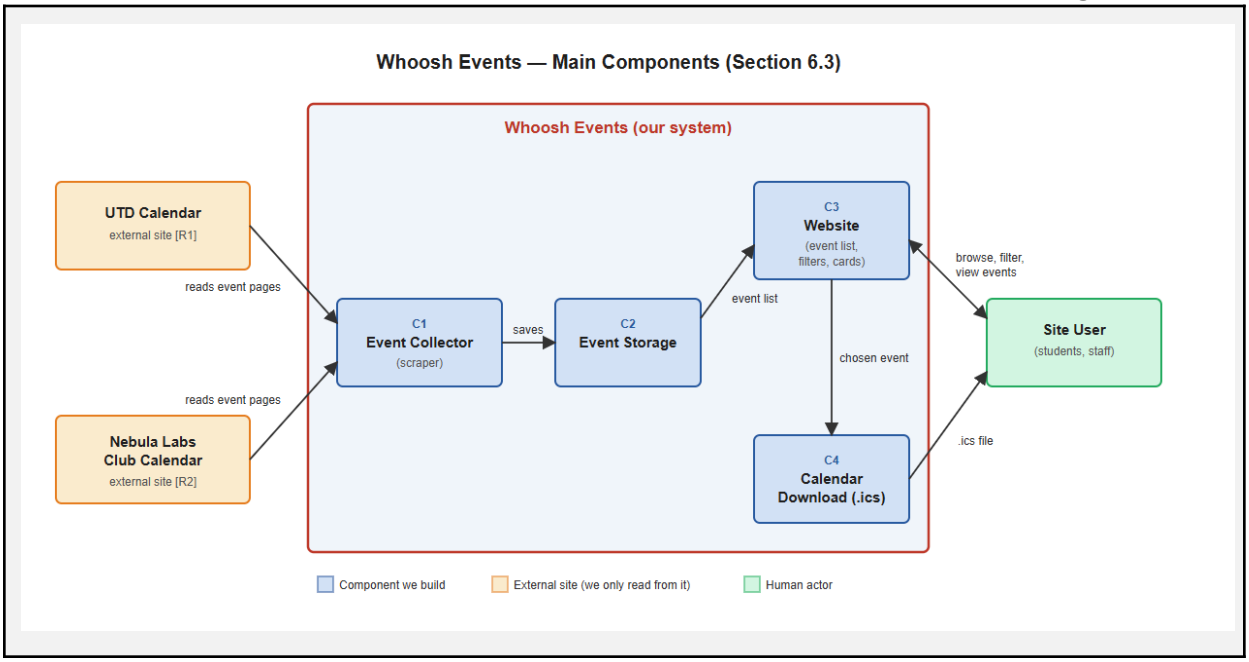


Figure 6.3 Component diagram: components, provided and required interfaces, external systems

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
C1	An event collector that runs on a specific timing and collects event information from other websites.	Duplicated event records	Viewable, public events from Nebula Labs’ Club Calendar and UTD Calendar.	UC-01, QR-03, QR-04, FR-03, FR-05, CON-02, CON-03, ADR-03

C2	Keeps records of the events taken from C1.	List of events that can be sorted and filtered.	Event details from C1	UC-01, QR-04, QR-06, ADR-03
C3	Website to display the information collected.	Displays the events	Events from C2, C4 .ics download	UC-01, QR-01, QR-02, QR-05, FR-01, FR-02, FR-04, FR-07 to FR-12, ADR-03
C4	A download that turns a selected event into a .ics file that a user can easily import to their calendar.	.ics file	Event Details from C2	UC-01

6.4 Deployment view

► OWNER: Austin · due Fri Nov 6, 11:59 PM (delete this line before submitting)

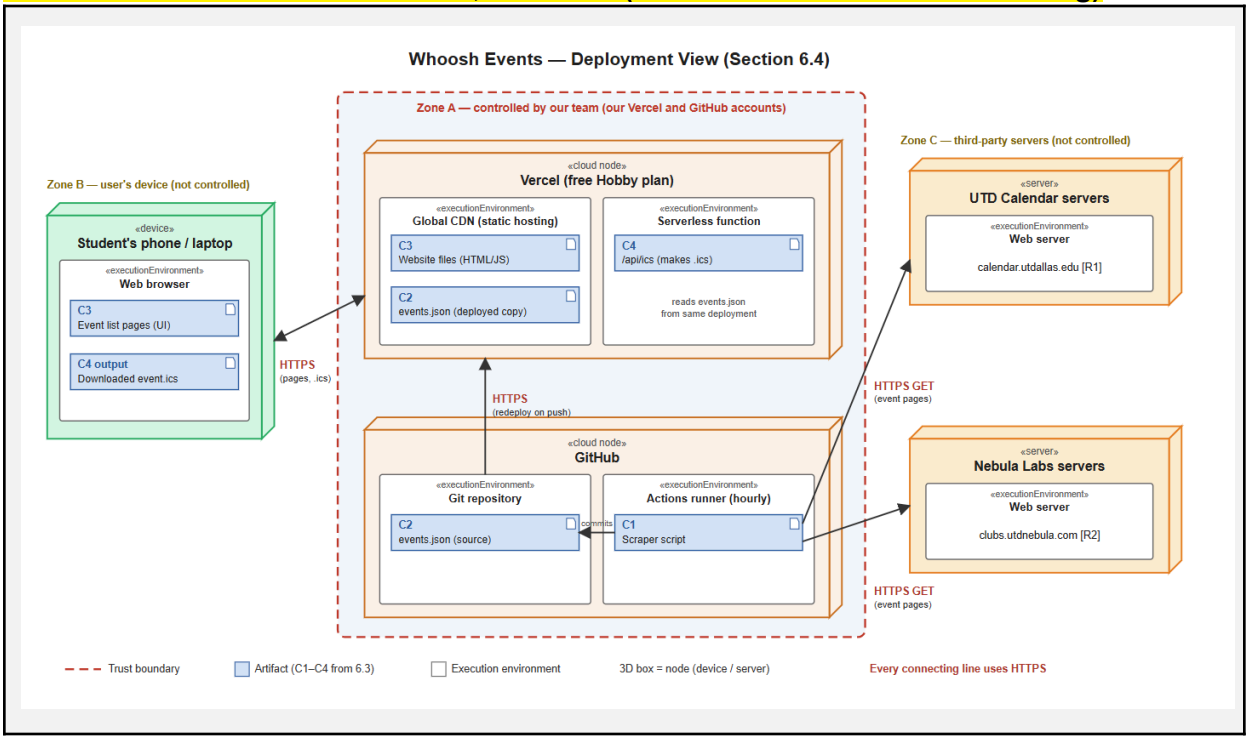


Figure 6.4 UML deployment diagram: nodes, execution environments, deployed artifacts, communication paths with protocols, trust boundaries

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
Student's Phone/laptop	A web browser page that lists all events and can receive a downloaded .ics file.	Vercel (HTTPS)	B: user device
Vercel	Global CDN that hosts our website files and a copy of events.json.	User's browser (HTTPS), GitHub	A: controlled by us
GitHub	A repository holding events.json and containing the actions runner that can execute our scraper.	UTD Calendar and Nebula Labs	A: controlled by us
UTD Calendar	calendar.utdallas.edu [R1]	Github Action Run	C: third party
Nebular Labs	clubs.utdnebula.com [R2]	GitHub Action Run	C: third party

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Every component in 6.3 is deployed somewhere in 6.4. Every communication path names its protocol. Mark the trust boundaries: where data crosses from a zone you control into one you do not.
If you use a cloud service, say which responsibilities stay with your team under the shared-responsibility model.
Budget: one page including the diagram.

6.5 From analysis to architecture

Analysis object (from 5.1 and 5.2)	Component that owns it (6.3)	Note (split, merged, external)
Event	C2 Event Storage	C1 creates and merges events. C3 and C4 only read them.

Tag	C2 Event Storage	Stored with the event. C1 assigns tags and C3 offers them as filters.
Organizer	C2 Event Storage	Stored with the event. C1 reads it from the source page.
SourceListing	C2 Event Storage	Saved with the event. C3 links to it.
SourceSite	C2 Event Storage	Split. The real sites (R1 UTD Calendar, R2 Nebula Labs) are external and we only read them. C2 keeps our record of each site: address, last checked time and last successful update time.
SharedLink	C3 Website	The link is created and opened in the website.
EventBrowser (interaction object)	C3 Website	Runs the user's visit and is in control. C3 is the only component the user talks to.
EventCatalog (interaction object)	C2 Event Storage	Merged with the stored Events and Tags. It is read access to C2, not a separate component.
LinkRegistry (interaction object)	C3 Website	Merged with SharedLink.

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Every class and interaction object from section 5 lands in exactly one component, or the note says why it was split, merged or left to an external system. This table is how the reader sees that the architecture realizes the analysis rather than replacing it. Budget: a quarter of a page.

6.6 Design principles applied

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Separation of concerns	Collecting events, storing them, showing them and making a calendar file are four separate jobs.	C1 Event Collector, C2 Event Storage, C3 Website, C4 Calendar Download
Information hiding	Only the collector knows how the UTD Calendar and Nebula pages are laid out. The website never reads a source site and only sees stored events.	6.3: only C1 connects to R1 and R2. 6.4: the website reads the deployed events.json only
Low coupling	The collector and the website share one file, events.json, and never call each other. If a source site changes, only C1 changes.	GitHub Actions writes events.json to the repository and Vercel redeploys. ADR-01
High cohesion	Each component does one thing. C4 only turns one chosen event into an .ics file. C3 only displays and filters events.	6.3: C4 takes the chosen event and returns an .ics file
Design for change	A third event source would be a new reader inside C1, with nothing else changing. The refresh interval and the host are recorded decisions, so they can be swapped.	6.3: C1 is the only component tied to a source. ADR-02, ADR-03
Keep it simple (KISS)	No accounts, no database server and no user-created events. One scheduled job, one JSON file and a static website on a free plan.	6.4: Vercel and GitHub Actions only. 1.2: out of scope list

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
One line per principle, pointing at something concrete. A principle you cannot point at in your own design should be left blank and explained in one sentence, not filled with a definition from the slides. Budget: a third of a page.

7 Design Rationale

7.1 Architecture decision records

► OWNERS: one per ADR (see each ADR below) (delete this line before submitting)
At least three. ADR-01 records the style decision from 6.2. Use the format you practiced in the Lesson 16 workshop, written about your own project. Copy the block for each decision.

ADR-01 Architectural style

► OWNER: Habib · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Status	Accepted / Superseded by ADR-
Date and owner	11/06/2026
Context (drivers)	Which QAS, use cases and constraints force a decision here
Options considered	1. Repository style: a scheduled collector writes events.json and the website reads it. 2. Client-server with live fetch from the source sites on every page view. 3. Event-driven or microservices with a message queue.
Decision	Use the repository style . C1 runs on a schedule and writes events.json. C3 reads only that file . C1 and C3 never call each other.
Consequences	Name the quality it favors, the move, and the cost you accept
Evidence	Spike result (section 8), measurement, documentation, or "none yet" with an open question
Related	ADR-02, ADR-03, QAS-01, QAS-02, QAS-03, R1

ADR-02 Event refresh strategy

► OWNER: Jaiom · due Fri Nov 6, 11:59 PM (uses the spike numbers) (delete this line before submitting)

Status	Accepted / Superseded by ADR-
Date and owner	
Context (drivers)	Which QAS, use cases and constraints force a decision here

Options considered	At least two, each in one line
Decision	
Consequences	Name the quality it favors, the move, and the cost you accept
Evidence	Spike result (section 8), measurement, documentation, or "none yet" with an open question
Related	ADR-, QAS-, R- (risk), CH- (change)

ADR-03 Hosting platform

► OWNER: Austin · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Status	Accepted
Date and owner	11/4/2026 | Austin Ou Yang
Context (drivers)	CON-04 states that we need free hosting for our site. In section 10.4, we also state that we need to be able to maintain separate branches (production and preproduction)
Options considered	1. Vercel - Can allow preview branches, easy deploys through git, and is able to host for free. 2. Netlify - Similar to git, but has less free features and slower support for our framework features. 3. Cloudflare Pages - Has the most free bandwidth, but takes a lot more time to setup.
Decision	Vercel will be used to host our Whoosh Events website.
Consequences	This choice is in favor of QR-01 and CON-04. Move: We connect the Git repo to Vercel, where production(main) connects to the live site and preproduction goes to the preview site. Cost: The Vercel Hobby plan has limits on scheduled jobs and may not allow us to run our scraper hourly.
Evidence	none yet. Open Question: does Vercel Hobby (free plan) allow our scraper to be run at least every 2 hours?
Related	CON-04, QR-01, QR-03, QR-06

7.2 Trade-off summary

Quality favored	Architectural move	Cost accepted	Recorded in
			ADR-1
			ADR-2
Low cost (CON-04) and fast page load (QR-01)	Host the website on Vercel, connected to the Git repository. The main branch is the live site and the preproduction branch is the preview site.	The free Vercel Hobby plan limits scheduled jobs, so it may not run the scraper every hour or two.	ADR-3

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
One row per real trade-off. This is the table a reviewer reads first. Budget: a quarter of a page.

8 Prototype Evidence: Architecture Spike

► OWNER: Jaiom · due Fri Nov 6, 11:59 PM (code + README + numbers in /spikes/S-01) (delete this line before submitting)

Spike ID and title	S-01
Risk or decision it addresses	R- / ADR- / QAS-
Question (answerable yes or no)	
Set-up	Stack, data, what was simulated; folder in the repository
How it was measured	
Result (numbers)	
What it changed or confirmed	Which ADR, QAS or requirement
Where to see it	Commit hash, /spikes/S-01/, screenshot or log file

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Build the smallest thing that answers your riskiest architectural question, run it, and measure it. Time-box it to a few hours. It may be thrown away afterwards; it is not the product.
Good spikes answer questions such as: can we read the external data source, and is it fresh to within 60 seconds? Does the chosen hosting meet QAS-01 under a simulated peak? Does the authentication provider fit our trust boundary?
Never commit passwords, API keys or other credentials; keep them in local configuration excluded from the repository.
The code, the measurement and a short README are committed to /spikes before the deadline. A spike described but not in the repository is not evidence.
Budget: half a page.

9 Traceability

9.1 Extended traceability matrix

► OWNER: Atharva · due Sat Nov 7, 6 PM (delete this line before submitting)

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-01	UC-01	Event, SourceListing	C2, C3	ADR-01, ADR-03	S-01
FR-02	UC-01	Event, Tag	C3	ADR-01, ADR-03	None yet (website development in progress)
FR-03	UC-03	Event, SourceListing, SourceSite	C1, C2	ADR-02, ADR-03	None yet (website development in progress)
FR-04	UC-06	SourceSite	C3	ADR-03	None yet (website development in progress)
FR-05	UC-01, UC-05	Event, SourceListing, SourceSite	C1, C2, C3	ADR-02, ADR-03	None yet (website development in progress)

FR-06	UC-04	Event	C3, C4	ADR-01, ADR-03	None yet (website development in progress)
FR-07	UC-01, UC-02, UC-03	Event, Organizer, Tag, SourceListing	C3	ADR-01, ADR-03	None yet (website development in progress)
FR-08	UC-05	Event, Organizer	C3	ADR-01, ADR-03	None yet (website development in progress)
FR-09	UC-02, UC-03	Event, SourceListing	C3	ADR-01, ADR-03	None yet (website development in progress)
FR-10	UC-02	SharedLink, Event	C3	ADR-01, ADR-03	None yet (website development in progress)
FR-11	UC-01	Event	C2, C3	ADR-01, ADR-03	None yet (website development in progress)
FR-12	UC-02, UC-03	Event, SourceSite, SourceListing	C3	ADR-01, ADR-03	None yet (website development in progress)
QR-01	UC-01	Event	C3	QAS-01, ADR-01, ADR-03	None yet (website development in progress)
QR-02	UC-01	Event, Tag	C3	QAS-01, ADR-01, ADR-03	None yet (website development in progress)
QR-03	UC-01, UC-02	Event, SourceSite	C1	ADR-02	None yet (website development in progress)
QR-04	UC-01, UC-02, UC-03	Event, SourceListing	C1, C2	ADR-02	None yet (website development in progress)

QR-05	UC-03	Event	C3	ADR-01	None yet (website development in progress)
QR-06	UC-01	Event, SourceSite, SourceListing	C2	ADR-02, ADR-03	None yet (website development in progress)
CON-01			C3	ADR-01	None yet (website development in progress)
CON-02	UC-01, UC-03	SourceSite, SourceListing	C1	ADR-02	None yet (website development in progress)
CON-03	UC-01	SourceSite	C1	ADR-02	None yet (website development in progress)
CON-04			C1, C2, C3, C4	ADR-03	None yet (website development in progress)

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Extend the P1 matrix one column at a time: from each requirement to the use case, to the analysis classes that realize it, to the component that will own it, to the decision or scenario that shaped it, and to whatever evidence exists (spike, measurement).
Requirements realized by the use cases you analyzed need the full chain. Every other MUST requirement needs at least its component. Withdrawn requirements stay in the matrix, marked as such.
Budget: one page.

9.2 Gap analysis

► OWNER: Atharva · due Sat Nov 7, 6 PM (delete this line before submitting)

Gap	Items	What you did about it
Requirements with no component	N/A (All requirements have at least 1 component)	

Components with no requirement, QAS or ADR	N/A (All components have a requirement, QAS, or ADR)	
Use cases not analyzed (and why that is acceptable)	N/A (Each UC corresponds to at least 1 requirement)	

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
An empty gap table is believable only if the matrix above is complete. Say what each gap means and what you did. Budget: a quarter of a page.

10 Project Plan (updated; was P1 section 4)

10.1 Roles and working agreement

Name	P1 Role	P2 Responsibility	Decision rule
Austin Ou Yang	Assist all sections	Repository owner and submitter. Owns 6.3, 6.4, ADR-03, 10.4 and the Vercel and GitHub setup. Submits the final PDF on Canvas.	Unchanged
Jaiom Rola	Sections 4 and 5, help with 1	S-01 spike (section 8), ADR-02, 10.3 risks and 11 open questions.	Unchanged
Ahmed Fouad	Section 3	Sections 4, 5.1 and 5.2 (use cases, domain model, sequence diagrams), 0.2 and the P1 fixes in 1 to 3	Unchanged
Sinan Can Demir	Section 4	6.5, 6.6, 7.2, 10.1, 10.2, 10.5 and 12 (signatures). Reviews Jaiom's spike PR.	Unchanged

Habib Mashhood	Section 2 and parts of 1	6.1 and 6.2 and ADR-01 (architecture decisions)	Unchanged
Atharva Thombare	Section 2	5.3, 9.1 and 9.2 (consistency check, traceability, gap analysis)	Unchanged

10.2 Plan to P3

Milestone	Date	Owner	Done when
UTD Calendar collector	Nov 20	Jaiom	FR-05 is met for UTD Calendar: 10 upcoming events picked from the source all appear in our event list
Nebula collector with last-good-data fallback	Nov 20	Habib	FR-05 is met for Nebula, and FR-03 is met: 3 events listed on both sources each appear exactly once. Old events stay if the site is down.
Scheduled collection job and Vercel deploy	Nov 22	Austin	The job writes events.json and Vercel redeploys on its own
Event list page with pagination	Nov 24	Ahmed	FR-01 is met (20 or more events counted on one page) and FR-11 is met (the next 20 load at the end of the list)
Filters	Nov 27	Atharva	FR-02 is met: for 5 different tags, every event shown has the tag and no upcoming event with it is missing. FR-08 is met for 3 organizers.
Details page, share link and .ics download	Nov 29	Sinan	FR-09 is met (5 event cards show the source description), FR-10 is met (5 copied links

			open the same event) and FR-06 is met (the .ics file downloads)
Automated tests	Dec 2	Jaiom	Collector and filter tests pass
Mobile and keyboard check	Dec 4	Atharva	Defects found are fixed and rechecked
"How to host an event" page	Dec 4	Ahmed	FR-04 is met: one working link per source site (2) to its official event submission page
Integration check of the whole site	Dec 6	Habib	The site works end to end on preproduction
P3 document checked against P2	Dec 7	Sinan	Every change to P2 is recorded in 0.2
Final review and submission	Dec 8	Austin	P3 is submitted on Canvas

10.3 Risks (updated)

► OWNER: Jaiom · due Fri Nov 6, 11:59 PM (start from P1 section 4.3) (delete this line before submitting)

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R1			
R2			
R6 (new)			

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Close or downgrade P1 risks with a reason. Add the architectural risks your design created, and point to the spike or ADR that addresses each.

10.4 Repository

► OWNER: Austin · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Public Repository URL: https://github.com/AO2-Git/whoosh-events
Our folder for p2 is layered like this: /docs/p2, /docs/p2/diagrams, and our spikes our located in /tests/spikes. Our branch and pull request rule is that every person makes their own branch for p2, then creates a pull request to merge into preproduction, then once everything is finalized in preproduction, we issue a pull request to main.

10.5 Use of AI tools in P2

► OWNER: Sinan collects everyone's rows · due Sat Nov 7, 6 PM (delete this line before submitting)

Member	Tool	What it was used for	What you checked or changed
Austin Ou Yang	Claude	Generated the 6.3 component diagram and 6.4 deployment diagram; drafted the 6.3 component table, the 6.4 node table and shared-responsibility paragraph, and the ADR-03 hosting decision record; compared Vercel with Netlify and Cloudflare Pages	Checked every component, requirement ID (FR/QR/CON) and source against P1 sections 1.2, 1.4, 2.1 and 3. Cross-checked 6.3 against 6.4. This caught two errors that were then fixed: a missing protocol label on one 6.4 line, and a missing C2 → C4 data flow in 6.3. Hosting claims were compared against the cited web sources.
Atharva Thombare	Google Notebook (Gemini)	***Section 9.1***: Drafted the extended traceability matrix. ***Section 5.3:*** Generated the list of items/objects for easier readability. Formatted the .md files for my commits	***Section 9.1:*** Checked each entry of the table with the respected section to verify the correctness. Made some changes to the table that was drafted due to it missing some components when cross-referenced. ***Section 5.3:*** Checked each generated item/object that needed to be cross-checked and noted whether each check was a pass/fail. The Ai model was not firm in its response to the pass/fail, so that part was disregarded and manually done.
Sinan Can Demir	Claude Code	Drafted sections 6.5, 6.6, 7.2, 10.1, 10.2 and	I compared every row of 6.5 against the classes in 5.1, the interaction objects in

	(Sonnet 5.5)	10.5 and my row in 0.1 (committed as d0935fd, 01ca14e and cd4ccbd in PR #6). Helped me read teammates' PRs (#2, #4 and #5, including the S-01 spike code), check my sections against the template guidance and the working document, and draft my review comments on PR #2 and PR #5, including the approval on PR #5.	5.2 and the components in 6.3, and had it rewritten so each object has one owning component, with notes for the split and merged cases. I rebuilt 10.2 with the template's four columns and tied each "Done when" to the acceptance criteria in 3.1. I checked 10.1 against my P1 section 4.1 and the team's work. I checked the 6.6 evidence against 6.3 and 6.4. I read and approved each review comment before posting it from my own account; the PR #5 approval was posted as drafted. I left the 7.2 cells that depend on ADR-01 and ADR-02 blank until those ADRs existed.
Ahmed Fouad			
Habib Mashhood	Claude	Drafted 6.1 (QAS-01 to QAS-03), the 6.2 style table and paragraph, ADR-01, and my row of the 7.2 trade-off summary from our P1 requirements and the 6.3 and 6.4 sections.	Checked each QAS number against QR-01, QR-03 and QR-06 in section 3.2. Checked that the style choice matches the components in 6.3 and the nodes in 6.4. Compared ADR-01 with ADR-03 for consistency.
Jaiom Rola			

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Declare every use, including code assistants used in the spike. Commit messages carrying an AI co-author line must be matched by a row here. Using AI is permitted; using it without declaring it is not. Write "None" if no tool was used.

11 Open Questions (updated; was P1 section 5)

► OWNER: Jaiom · due Fri Nov 6, 11:59 PM (start from P1 section 5) (delete this line before submitting)

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q					
Q					

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Close the P1 questions you answered and say how. Add the questions analysis and design raised: a step whose triviality was unclear, a scenario with no agreed response measure. If no questions remain open, say how the earlier ones were resolved.

12 Baseline v1.0 Declaration (was P1 section 6)

► OWNER: Sinan collects all 6 signatures · due Sat Nov 7, 6 PM (delete this line before submitting)
We declare this document, version 1.0, the baseline for P2. Sections 1 to 9 describe what we will build in P3. Any later change to them is recorded in section 0.2 with its reason.

Team member	Signature or initials	Date
Sinan Can Demir	SCD	11/08/2026
Austin Ou Yang		11/08/2026
Atharva Thombare		11/08/2026
Ahmed Fouad		11/08/2026
Habib Mashhood	HM	11/08/2026
Jaiom Rola		11/08/2026

How P2 Is Marked
Delete this page and the next before submitting. 60 points: 48 for the shared team artifact (80%) and 12 for individual contribution evidence (20%).

Criterion	Points	What full marks look like

Baseline, change control, plan and repository (sections 0.2, 1 to 3, 10 to 12)	6	Every point of the P1 feedback is answered in 0.2. Requirement IDs are preserved and withdrawn items marked. The plan to P3, risks, AI-use declaration, open questions and signed v1.0 baseline are up to date. The repository is accessible and, before the deadline, holds this PDF, the diagram sources and the spike, with the tag p2 on that commit.
Use-case analysis (section 4)	8	At least three use cases expanded turn by turn with step (0), one turn per row, an acknowledgment at the end, real alternate flows, and no technology. Preconditions are true assumptions, not hidden alternate flows. The self-review log records what each check found, including any defects fixed.
Domain model and object interaction (section 5)	10	A conceptual domain model with attributes and multiplicities that contains every item the actor columns mention. At least two use cases carried through non-trivial steps, a five-column scenario table and an analysis sequence diagram whose messages match the table row for row. The six cross-checks are run honestly.
High-level design (section 6)	10	At least three quality-attribute scenarios with response measures. A named style chosen over at least two alternatives for reasons tied to those scenarios. A component view in which every component answers to a use case, scenario or decision; a UML deployment view with protocols and trust boundaries; every analysis object mapped to a component; design principles pointed out in the design itself.
Design rationale (section 7)	6	At least three decision records, each with real alternatives, a decision and a stated cost. The trade-off summary agrees with them.
Prototype evidence (section 8)	4	One spike that answers a real architectural question with a measured result, committed with its README before the deadline, and linked to the decision it changed or confirmed.
Traceability (section 9)	4	Every requirement realized by the analyzed use cases is traced through analysis classes to a component, and to the scenario or decision that shaped it where one exists. Every other MUST requirement reaches at least a component. Gaps are named, explained and acted on.
Document quality	up to −3	Deducted from the team points, not awarded: guidance, example or warning boxes left in, placeholders, sections out of order or

		renumbered, unreadable diagrams, or a document far beyond its length budget.
Individual contribution evidence (section 0.1)	12 each	12: owns specific sections or artifacts, with own commits or pull requests before the deadline and at least one review of a teammate's pull request. 9: specific ownership and own commits, but thin or no review evidence. 6: specific claim with indirect evidence only (issues owned, or work uploaded explicitly on the member's behalf). 4: specific claim with no repository evidence. 2: vague or duplicated claim. 0: no credited work. A member with no credited work at all (absent from the contribution record and with no commits) may also lose the team portion, because the syllabus does not allow credit for work completed by others. This applies to individual non-participation only, never to a whole team.

WHERE P1 MARKS WERE LOST (DELETE THIS BOX BEFORE SUBMITTING)
Repositories that were empty, unreachable, or held a single upload from one account at the deadline. Repository and contribution credit could not be given, however good the document was.
Contribution records that cited commits, pull requests or files that were not in the repository. Everything cited is checked.
Guidance boxes, placeholders and unconfirmed draft text left in the submitted document.
Several members uploading different versions to Canvas. One member submits. If more than one file is uploaded, the latest upload is the one graded.
Quality requirements with no number in them, and requirements whose source was still "pending".
Work committed after the deadline. It is not counted, even a few minutes after.